Clinical Practice in Pediatric Psychology

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ISSN: 2169-4826

2022, Vol. 10, No. 4, 452–461
https://doi.org/10.1037/cpp0000463

Electronic Monitoring of Adherence in Clinical Care: Ethical and Practical Considerations

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Objective: Adherence to prescribed medical regimens is a key behavior for managing disease burden and progression. Despite the importance of understanding a child's adherence, many evidence-based assessment strategies have not been implemented into subspecialty clinical practice. The aim of this article was to highlight the potential ethical and regulatory challenges of using electronic monitoring (EM) of adherence in clinical care and offer practical solutions. ***Methods:*** Relevant ethical and legal standards were researched and summarized as applicable to EM of adherence. ***Results:*** Issues related to Human Relations, Privacy and Confidentiality, Record Keeping, and Assessment all apply to EM of adherence. ***Conclusions:*** Clinicians should be mindful and seek to proactively address issues related to informed consent, autonomy, privacy, and equity before starting an EM program of adherence. Multidisciplinary teams should thoroughly consider the ethical and legal aspects of implementing this form of adherence assessment into clinical practice.

Implications for Impact Statement
This topical review is the first to describe the practical implications of utilizing electronic monitoring of adherence in a clinical care setting with a multidisciplinary medical team. This is an important reference for any pediatric psychologist or practitioner who wants to be fully informed about the potential ethical and legal ramifications of using this technology with youth.

Keywords: adherence, electronic monitoring, ethics

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Adherence to prescribed regimens has been a focus area in pediatric psychology because it is applicable to the management of most childhood chronic medical conditions. This topic has become even more important as once fatal diseases (e.g., cystic fibrosis, cancer) have become more chronic in nature with scientific advancements and the increasing availability of new efficacious treatments, including oral, injectable, and inhaled medications. Given the impact that suboptimal adherence can have on morbidity, mortality, quality of life, health care utilization, and health care costs, improving adherence and self-management behaviors is critical for the field (Modi & Driscoll, 2020).

To intervene and improve adherence, clinicians must be able to assess it easily and reliably. A multitude of adherence assessment methods have been used across studies, including self-report, electronic monitoring (EM), laboratory markers,

We have no conflicts of interest to disclose. No funding was received.

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and pharmacy refill data. The consensus among researchers has highlighted the value of combining multiple adherence assessment methods (ideally, objective, and subjective) as an individual measure is unlikely to yield perfect data (Hommel et al., 2017). Electronic monitoring of adherence, a more objective measurement tool, has been gaining popularity in research studies and often serves as the gold standard given relative advantages (Ingerski et al., 2011). In fact, technology now exists (with associated EM devices) to measure adherence to not only medications in pill form, but also inhaled medications, insulin delivery devices, and nebulized medications. Electronic monitors for oral medications typically take the form of a pill bottle (e.g., MEMS Cap, AdhereTech, and eCAPS) or pillbox (e.g., SimpleMed+) that records a time and date stamp when opened. This data are then used to calculate a rate of adherence based on how many times the device was opened relative to how many times it should be opened per the prescriber's instructions. Like all adherence assessment tools, there are relative advantages and disadvantages to EM. Some advantages include detailed data on adherence patterns, lack of reliance on patient recall/memory, and limited extra effort on behalf of the individual, while some disadvantages include expense, technology issues interfering with data transmission, and inability to confirm that the medication was actually ingested (McGrady et al., 2018; Riekert & Rand, 2002).

Electronic monitor data are often used as an outcome measure for randomized clinical trials focused on improving adherence (Foster et al., 2018; Greenley et al., 2015). In research settings, there is an informed consent and assent process, personnel support, and rigorous protocols related to the use of EM This is in contrast to less resourced settings (e.g., clinical practice) where adherence may not be routinely assessed at all or with methods (e.g., patient or provider report) that are typically inflated or erroneous (Duncan et al., 2014). Several factors contribute to increased EM use in the context of research versus routine clinical care. First, EM can be costly for the initial purchase price of the device, as well as ongoing monthly monitoring fees, akin to a cellular phone plan (Anghel et al., 2019; Choudhry et al., 2022, McGrady et al., 2018). Second, a significant amount of data are generated by EM (up to multiple daily data points) that needs to be viewed, managed, and analyzed (McGrady & Ramsey, 2020). Third, physicians and members of the clinical team are typically very busy with routine care and may have limited time or resources to implement an adherence monitoring program. Fourth, practitioners and patients alike may be hesitant to utilize EM due to the perceived invasiveness of monitoring, as well as ethical and legal concerns. They may be hesitant to have access to data of this kind and question their liability should suboptimal adherence result in an adverse outcome for a child or adolescent.

While these barriers reflect challenges in EM adherence measurement in clinical practice, there is equal concern regarding clinical decision making without this critical information. Clinical decisions (e.g., medication dose increases, adding additional medications to treat persistent symptoms, etc.) are often made with the assumption that a patient is following their prescribed treatment (Modi et al., 2012). Additionally, adherence can be a modifiable target for symptom management, reduced health care utilization, and improved outcomes. This is evident in published randomized clinical trials data demonstrating improved adherence with provision of EM data or feedback plus reminders (Morton et al., 2017; Sulaiman et al., 2018).

Opportunity for Adherence EMs in Clinical Practice

Despite the challenges noted above, there is an opportunity to incorporate EM and add significant value to patient care, medical decision making, and outcomes (Cortina et al., 2013; Hilliard et al., 2011). Herzer and colleagues (Herzer et al., 2012) presented an excellent review of the benefits of using EM in clinical care, including the ability to identify patterns of adherence problems that were specific to psychologists providing behavioral intervention. Further, one study in African American adults with hypertension found that individuals had positive perceptions of EM use for clinical care, including helping them to remember to take their oral medications (Schoenthaler & Ogedegbe, 2008). Despite these advantages, pediatric psychologists may need to partner with their medical colleagues to incorporate EM more broadly to patients who may lack access to a psychologist, which is especially salient to underserved populations (e.g., people belonging to racial/ethnic minority groups, individuals with public insurance, etc.). Additionally, the potential ethical challenges with EM were not reviewed by Herzer and colleagues (Herzer et al., 2012). A 2016 opinion paper explored ethical issues related to EM of

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medication adherence (Campbell et al., 2016). While this article presented a discussion on issues relevant to autonomy (the patient's right to determine what will be done to them and what happens with that information), privacy (the notion that personal information about the patients will not be divulged), and trust (confidence in the character of the provider; Rand & Sevick, 2000), the content was relevant to adult patients and did not explore considerations unique to children, including children's lack of capacity to consent to EM and the complexities of multiple individuals involved with adherence (e.g., patient, caregivers, etc.). Psychologists and their partnering medical colleagues would likely benefit from a topical review of the applicable ethical and legal considerations for EM in clinical care. Access to some of this information, along with potential solutions, may increase the likelihood that this evidence-based assessment tool reaches clinical practice.

Thus, the objective of this article is to describe the use of EM for the measurement of adherence in a pediatric population. Ethical issues, legal considerations, and practical tips for overcoming these issues will be discussed and facilitated by a case example. The cohort we chose was pediatric kidney transplant recipients receiving care from a large academic medical center in the Midwestern United States. Kidney transplantation is a treatment option for individuals with various congenital and acquired conditions causing kidney failure. Adherence is critical in this population, as transplant recipients must take immunosuppression medication at very specific time intervals for the rest of their life to prevent organ rejection. In addition, blood levels are closely followed to ensure there is an appropriate level of medication in the child's system to balance the benefits and risks of immunosuppression. Although kidney transplant will be the exemplar, the information and recommendations are broadly applicable to other chronic illness populations that must sustain a daily medication or treatment regimen. Given the nature of the present topical review, which does not include real patient data, this project was exempt from institutional review board and consent or assent procedures.

Clinical Case Example

"Joe" is a 16-year-old Black cisgender young man who underwent a deceased donor kidney transplant for end-stage kidney disease secondary to renal dysplasia 3 years prior. His initial course was uncomplicated. His maintenance medications included tacrolimus (immunosuppressant drug; must be taken exactly 12 hr apart every day) and extended-release nifedipine (antihypertensive drug; must be taken once per day). Recent laboratory data showed slowly increasing serum creatinine (blood marker of kidney function) and variable (including below target) levels of tacrolimus in Joe's blood work. The first interventions were to recommend aggressive hydration and increase the dose of tacrolimus. When neither of these interventions resulted in improvement, Joe's medical team began to consider whether he was missing doses of immunosuppressant medication.

As part of the kidney transplant program's adherence support program, Joe was completing a barriers assessment tool every 6 months during follow-up clinic visits. Neither Joe nor his mother endorsed any barriers to adherence on the last two administrations of the tool. The medical team began to discuss the concerns about rising creatinine and variable drug levels and their suspicion of possible adherence problems with Joe and his mother in clinic; however, they both believed that Joe was doing a good job taking his medication every day. The medical team discussed the availability of EM with Joe and his mother.

Application to APA Ethical Standards and Legal or Regulatory Considerations

The American Psychological Association's (APA) Ethical Principles for Psychologists and Code of Conduct provide one rubric from which ethical concerns and decisions regarding the use of EM can be considered. Standards 3 (Human Relations), 4 (Privacy and Confidentiality), 6 (Record Keeping and Fees), and 9 (Assessment) of the APA Ethics Code are particularly pertinent to using EM in clinical practice (see Table 1 for a summary and description). Viewing the issues of privacy, autonomy, and equity through this lens can provide guidance to practitioners currently using these tools, as well as inform the need for changes in existing codes and regulations to ensure protection for the patients and professional users. The use of adherence EM must also be considered in the context of legal and regulatory issues within an institution. Most companies who make adherence EMs are private and for-profit.

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Table 1
Summary of Applicable American Psychological Association Ethical Standards

Standard	Description
3 (Human Relations)	3.01 Unfair Discrimination
	3.04 Avoiding Harm
	3.07 Third-Party Requests for Services
	3.09 Cooperation with Other Professionals
	3.10 Informed Consent
	3.12 Interruption of Psychological Services
4 (Privacy and Confidentiality)	4.01 Maintaining Confidentiality
	4.02 Discussing the Limits of Confidentiality
	4.04 Minimizing Intrusions on Privacy
	4.07 Use of Confidential Information for Didactic or Other Purposes
6 (Record Keeping and Fees)	6.01 Documentation of Professional and Scientific Work and Maintenance of Records
	6.02 Maintenance, Dissemination, and Disposal of Confidential Records of Professional and Scientific Work
	6.04 Fees and Financial Arrangements
9 (Assessment)	9.03 Informed Consent in Assessments

Note. From *Ethical Principles of Psychologists and Code of Conduct*, by American Psychological Association, 2017 (https://www.apa.org/ethics/code/). Copyright 2017 by the American Psychological Association.

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It is essential that institutions have clear agreements with the adherence EM company related to how the data will be protected, stored, used, and shared.

Before Using EM With Patients

It is important to determine whether the EM and platforms are compliant with the Health Insurance Portability and Accountability Act of 1996, as amended (HIPAA) and to enter into a business associate agreement with the third party that operates the EM to ensure ongoing HIPAA compliance [Pub. Law 104-191, 110 Stat. 1936; 45 C.F.R. 164.504(e)]. Additional contractual protections may need to be considered if the EM vendor maintains or transmits adherence EM data outside the United States. Even with HIPAA privacy and security rules, security breaches can and do occur. One strategy for overcoming this is to have informed consent procedures around EM use so patients and caregivers are clear about potential risks to privacy and confidentiality. To facilitate autonomy, providers should (a) give an appropriate explanation of what the EM device does, including the data it collects, (b) gain individual's consent to use the EM to store their medications, and (c) match an EM that is consistent with the patient's preferences and best interests before implementing EM with a patient (Faisal et al., 2020). For the medical team, this includes providing well-defined parameters regarding who will have access to the raw EM data, guidelines regarding who can view the raw EM data, and what kind of training individuals need to have to view and interpret the raw EM data. Once interpreted, there should be clear guidelines regarding which medical team members the data will be shared with, how the data will be shared, and how the data could be used to inform their treatment (3.09 Cooperation with Other Professionals; Klugman et al., 2018). When the patient is school-aged or an adolescent, it is important that the team explain the EM device using developmentally appropriate terminology (e.g., avoiding jargon). Ultimately, EM may be something that parents opt for their children (akin to a new medication), but it is incumbent on the team to explain what this means for the minor using terminology they can understand (e.g., "This special box will let us know how it's going with taking your medicine so we can help you."). Information should be provided to the patient regarding what EM data will be included in the patient's permanent medical record (6.02 Maintenance, Dissemination, and Disposal of Confidential Records of Professional and Scientific Work). Finally, all of these discussions and the patient's response (i.e., oral consent or assent) should be appropriately documented in the patient's medical record (6.01 Documentation of Professional and Scientific Work and Maintenance of Records; 9.03 Informed Consent in Assessments).

In addition to ensuring the patient understands who on the medical team will have access to their data, providers should inform the patient if using the EM will require the involvement of a third party. Use of EM's typically require engaging a third party

company that owns the EM and extracts data to a storage cloud that can be viewed on an application (3.07 Third-Party Requests for Services). Additional questions to inquire about include whether the deidentified data will be sold to marketing or insurance companies to better understand adherence behaviors, or if the company will monetize the adherence data in some way. For example, a company could create benchmarks of adherence for the industry or share the data with particular pharmaceutical companies for profit. Data uses of this type may require specific patient consent under HIPAA [45 C.F.R. 164.508(a)(3)-(4)].

Providers are responsible for only recommending devices that are HIPAA compliant and have been vetted for EM reliability and accuracy. While companies often provide their internal data regarding reliability and validity, few studies have been conducted externally to ensure the EM timestamp and recording of doses are correct, the troubleshooting of technological failures are identified, and the data are fully accurate. McGrady et al. (2018) conducted an independent evaluation of various adherence EM with key benchmarks identified to demonstrate the internal and external validity. Unfortunately, several companies were unwilling to provide their products for evaluation. It is incumbent upon the clinical team to ensure that the monitors are reliable to ensure that patient-facing information, which will be acted upon clinically, is based on the most robust and reliable information available. Consistent with these guidelines, Joe and his mother agreed to use an electronic pillbox (as Joe was already accustomed to organizing his medications in this manner), SimpleMed+, for a few months to determine if adherence may be playing a role in the lab values seen on bloodwork. This device was vetted by the medical team after extensive research on EMs on the market. In addition, ensuring that patients are informed not only that a third party has access to the data, but that despite robust data security and privacy protections, as with any digital storage, there is an inherent risk of data compromise. Next, providers should help patients and their caregivers obtain assistance to understand the parameters of the user agreement that they will be making with the EM company (3.10 Informed Consent). Ensuring the materials are presented in patient-friendly terms and that match the health literacy of families is critical. In the case of Joe, any terms that were technological in nature (e.g., data transmission, cloud, etc.) were explained. For example, we explained to the family that there was a chance that someone could see their data. Additionally, helping patients understand the costs and potential reimbursement through insurance for monitoring medication adherence is important. While very few insurance companies have been reimbursing the use of EM (i.e., use of insulin pumps and blood glucose monitors for Type 1 diabetes), an expansion of reimbursement is anticipated. In 2021, the Centers for Medicare and Medicaid Services (CMS) approved reimbursement for remote therapeutic monitoring (86 Fed. Reg. 64996, 65115-7, Nov. 19, 2021). In doing so, CMS announced that these codes (CPT codes 98975, 98980, and 98981) could be used for monitoring medication adherence. While these codes are currently limited to monitoring respiratory and musculoskeletal systems, Medicare reimbursement trends suggest reimbursement for additional bodily systems, and possibly mental health, will be coming soon.

The team must discuss how the adherence data captured will be used in clinical care and what responsibility the clinical team versus the company have when a patient is not adhering to treatment. Third party passive monitoring or lack of monitoring on the part of the company is likely common. The clinical team should establish key benchmarks around definitions of significant nonadherence that warrants clinical attention. For example, within our own kidney transplant center, clinical consensus was that patients should be contacted if they missed taking their immunosuppression medication for 48 hr. This number was felt to be conservative to ensure the safety of the patient. Furthermore, we assigned a clinical team member to review and respond to adherence data to ensure all patients were being appropriately followed. As part of the informed consent and assent process, it is critical that this is reviewed with caregivers and the patient. Specifically, adolescents should be informed that their designated caregiver will be contacted if there is a lack of activity on the EM (suggesting missed doses), as this may create some conflict with the caregiver-child relationship, especially if the caregiver presumed the patient was adherent. While discussing this with Joe and his mother, the medical team encouraged them to think in advance about how they would handle these calls if received. For example, Joe's mother promised to ask Joe what was going on first, instead of assuming that he was just not taking his medication.

If there will be any additional cost associated with the EM, this should be described to the patient as part of the consent process before prescribing

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EM use (6.04 Fees and Financial Arrangements). The use of EM in clinical practice is early in development and it may not be clear to patients and families that EM devices and accompanying applications can have a significant cost. Compounding this issue is that, currently, the majority of health systems do not have standing supplier contracts with EM companies and the cost of EM and its use will likely not be integrated into the medical billing system. This places an additional responsibility on providers to ensure that any potential costs associated with being prescribed an EM are clear to the family. In addition, providers must be prepared to manage any limitations in providing EM if it is dependent on the family's ability to pay. This includes what the provider intends to do if the family does not pay for the services (e.g., sending the family to a collection agency) or devices that are "loaned" become lost. It is important to balance health equity with the beneficiary inducement prohibition in the Civil Monetary Penalty Laws, which generally prohibits offering free items and services to Medicare and Medicaid beneficiaries [42 U.S.C. §1320a-7a(a)(5)]. To achieve the proper balance, organizations should implement carefully developed policies and procedures regarding loaned or gifted devices. Joe's family was informed that there would not be any out-of-pocket costs to them or charges billed to their insurance company.

Taking into account patient preferences regarding the type of EM can facilitate use by matching the device to the medication regimen as well as by demonstrating respect for the patient's right to privacy. An intrusion on privacy may be as fundamental as how visible an EM is in one's home. A study among older adults tested EMs that ranged from small pillboxes to larger pillboxes and automated dispensers (Faisal et al., 2020). The majority of participants indicated that they disliked the notion of using large EMs and preferred EMs that were less noticeable (4.04 Minimizing Intrusions on Privacy). This may be particularly true for subgroups of patients who have experienced systemic racism or mistrust within hospital systems. Relatedly, children and adolescents may take their medication in different environments (e.g., splitting time between two caregivers, spend time grandparents' house, sleepover parties with friends, etc.). Before accepting the use of an EM, patients should be prompted to think about all of the places they may need to take the EM, and the extent to which they are comfortable with a visible device. All potential caregivers should be present for the conversation to ensure they are aware of EM, can think through logistics in advance, and know who to contact in case of technological problems.

The family received a brief demonstration on the pillbox and was provided instructions on when and who to contact on the clinical team for troubleshooting. Once all questions were answered to the family's satisfaction, the EM was activated by the clinical team and the family was ready to use the device once they returned home.

Considerations During EM Use

When using digital technology in patient care, how providers use and communicate information generated for the devices cannot only influence the accuracy of the assessment and the effectiveness of treatments, but can directly impact the patient-provider therapeutic relationship. Data from EMs can only be accurately interpreted with information provided by the patient and, when appropriate, their family members. Take for instance EM data that shows perfect medication adherence over 30 days. This could reflect true adherence behaviors, or alternatively, a family member opening the pillbox compartment to deactivate the reminder alarm each day. In the latter case, the provider is only going to glean this information in the context of a discussion in which the patient and family trust the provider and feel safe doing so. This example highlights how even misuse of an EM can provide an opportunity for the provider to build trust and facilitate conversation about what is the most helpful adherence support for the patient.

Another key consideration in using EM with patients relates to the record keeping of EM data. The majority of EMs do not interface with the electronic medical record. Therefore, it is incumbent upon providers to develop policies with regard to what EM data will be included in the medical record and inform the patient of how the EM information will appear (3.10 Informed Consent; 6.01 Documentation of Professional and Scientific Work and Maintenance of Records; 9.03 Informed Consent to Assessment). From a risk management perspective, it is important to determine how EM data will be reviewed to avoid having damaging data unknowingly residing in the medical record. The data should be viewed as any other health data and protected by HIPAA. If patient EM data are being used to teach other professionals, reasonable steps should be taken to protect and deidentify the

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information being shared unless the patient or guardian has consented to share protected health information for the purpose that it is being used (4.07 Use of Confidential Information for Didactic or Other Purpose).

About 3 weeks into Joe's use of the EM at his home, his transplant nurse coordinator was notified by the data monitoring team that there was no recorded activity on the EM for the past 3 days. Because that exceeded the predetermined threshold, the transplant coordinator contacted Joe's mother to inquire and troubleshoot the issue. She discovered that the family was traveling out of town for the long weekend and forgot the power cord, so they were advised to plug the EM in once they returned home. When Joe returned to his next clinic visit, his provider reviewed a calendar depiction of all on time, late, and missed immunosuppression doses. Of note, days where there were technical issues were grayed out. This facilitated a conversation on patterns of adherence, which gleaned that Joe missed taking his evening dose of tacrolimus five times more often compared with the morning dose. The provider utilized open-ended questions (e.g., "What do you notice about how you took medication last month?"; "What do you think will help make it easier to take medication twice per day?"). Problem-solving around barriers to evening medication adherence was initiated as a first step; however, Joe had an evening work shift that could not be changed. As a result, the provider suggested switching to a once daily form of immunosuppression (tacrolimus extended-release) that could be taken in the morning, to which the family agreed. Joe and mother were advised to continue EM to test the effectiveness of this intervention on adherence.

Terminating EM Use

Finally, the provider should outline a process for the patient to terminate use of the EM when they choose that includes a shared decision-making process information regarding the pros and cons of terminating EM use and potential implications, if any, for their treatment moving forward (3.10 Informed Consent). As EMs are used to support medication self-management, it is important that these discussions include information regarding the length of time the institution will be providing the EM (if applicable), conditions under which the use of the EM would be terminated, and options for the patient if they wish to continue (3.12 Interruption of Psychological Services). In addition, there are differences across EM companies regarding what happens to the data that was previously collected and the extent to which it can be deleted. Therefore, psychologists and medical providers should consider the data policies of the EM companies they are using with regard to deleting past data so they can reasonably inform patients of these policies before use (6.02 Maintenance, Dissemination, and Disposal of Confidential Records of Professional and Scientific Work).

Psychologists and their medical teams must remain cognizant of the potential for unfair discrimination (Standard 3.01) when using EM (Sim, 2019). This could take the form of using EM more or less often with a group of patients based on age, sex, gender identity, race, ethnicity, culture, national origin, religion, sexual orientation, disability, or socioeconomic status. Ensuring equitable access to the EM and applications (or cloud servers or portals) is critical for ethical EM use. Individuals who are older, have a lower income, live in a rural area, or have a disability are at greater risk of not having adequate digital access (Rainie, 2017). Although approximately 77% of adults in the United States have a smartphone (Smith, 2017), owners are more likely to have higher income (93% of adults who make greater than $75,000 per year own devices compared with 64% of adults who make less than $30,000 per year; Pew Research Center, 2017). Therefore, patients and families falling in a lower socioeconomic group are less likely to have access to a digital self-management tool, which further contributes to health disparities. If used for patient care, ensuring equal access to not only the EM device, but Internet access and smartphone devices to receive feedback from EM is incumbent on the psychologist and health care system within which they practice. Some authors have suggested creating technology loaner programs that provide this access for patients in lower socioeconomic statuses (Klugman et al., 2018). Other groups for whom use of EM could raise issues of equity include those who have lower health literacy and numeracy proficiency levels who may have greater difficulty interpreting the data provided by the application (Centers for Disease Control and Prevention, 2018). In this situation, providing a device will not resolve the challenges that EM presents. For these reasons, it is important to conduct a patient-centered assessment of potential barriers before considering EM as well as mitigate or eliminate the barriers to EM usability.

Joe's subsequent use of the EM demonstrated overall improved adherence in the following 2

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months, which was also reflected in his laboratory values (lower standard deviation of immunosuppressant drug levels, creatinine returning to baseline). Given sustained adherence over a couple of months, the family and medical team collectively determined that Joe no longer needed the EM as a clinical tool and it was returned to the clinical team. In some situations, the patient or family may be reluctant to discontinue EM use due to perceived benefits. In this case, the care team should work with the family on how to maintain good adherence without the EM, such as using a traditional pillbox and related behavioral strategies they can use with the nonelectronic pillbox (e.g., keeping it visible, pairing with a routine, etc.).

Conclusions

Electronic monitoring can be an excellent supplement to subjective adherence assessment methods in routine clinical care for many chronic health conditions. Given that daily monitoring generates a large volume of data, we recommend that the medical team collaborate with their center's Information Services (or equivalent) department to develop strategies for automating data collection and report generation that Risk Management approves. This will ease burden on the clinical team to cull through data or create reports that are easy to interpret. Newer EM companies are creating better interfaces that are provider and patient friendly, which should aid in this process. Additionally, most effectively utilize the data and provide transparency with patients and their caregivers, we recommend that are reports are generated to show trends and patterns in adherence data. There is clinical value in sharing adherence data with patients (similar to laboratory values and vital signs). If adherence is sufficient, then praise and reinforcement of existing strategies can be offered. However, if there are concerning trends of missed or late medication doses, then this facilitates conversation, identification of barriers, and implementation of evidence-based strategies to improve adherence. In fact, there is strong support for the efficacy of adherence promotion interventions by nonpsychologists (Wu & Pai, 2014) and our own team has demonstrated success in reducing organ rejection episodes when implementing a systematic approach to address adherence barriers in the clinic setting (Varnell et al., 2021). Clinicians looking for additional guidance on selecting EM technology that meets the needs and resources of their specific pediatric population can consult a recent review by Mason et al. (2022). In the future, consensus guidelines for integrating EM into clinical care should be generated and published.

Overall, incorporating EM into clinical practice will require additional time and other resources, as well as significant planning before implementation. However, there are significant benefits to be gained from having access to objective adherence data and incorporating it into usual patient care.

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Received March 31, 2022
Revision received September 26, 2022
Accepted September 29, 2022 ■

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Call for Nominations

The Publications and Communications (P&C) Board of the American Psychological Association has opened nominations for the editorships of *Clinician's Research Digest*; *Psychology, Public Policy, and Law*; *Psychology and Aging*; *Professional Psychology: Research and Practice*; *Journal of Experimental Psychology: Learning, Memory, and Cognition*; and the *Journal of Personality and Social Psychology: Interpersonal Relations and Group Processes*. Marisol Perez, PhD, Michael E. Lamb, PhD, Elizabeth A. L. Stine-Morrow, PhD, Kathi A. Borden, PhD, Aaron S. Benjamin, PhD, and Colin Wayne Leach, PhD, respectively, are the incumbent editors.

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